

Four-phase 1 MHz voltage regulator, 119 A

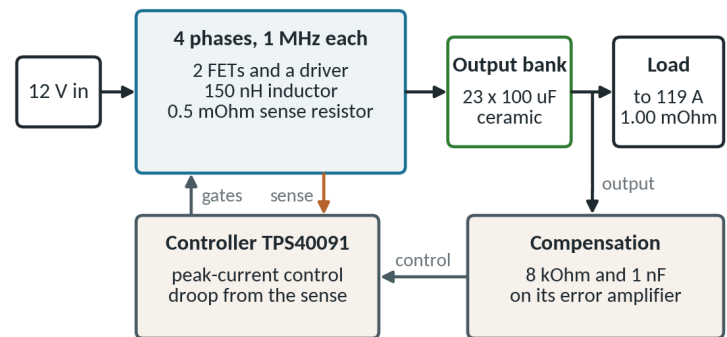
12 V in, VID 1.4 V on a 1.00 mOhm load line, all-ceramic output, for Intel VRD 10.1 (775_VR_CONFIG_04B)

PJ Crawley LLC, evidence case 2. Edition 2, 2026-10-04.

SIMULATION DATASHEET. NOT BUILT, NOT MEASURED.

What it is

- Four phases at 1 MHz each, the clocks 250 ns apart
- 12 V in (10.2 to 13.8 V), 0 to 119 A out on a 1.00 mOhm load line
- All-ceramic output bank: 23 x 100 uF
- Controller TPS40091: peak-current control, the droop from a 0.5 mOhm sense resistor in each phase
- Held to Intel's VRD 10.1 design guide, line by line, in simulation
- Runs with no load; follows VID from 0.8375 V to 1.6 V with an ideal modulator (the TPS40091 has no VID input)
- Loop crossover under 200 kHz with 75 deg or more of phase margin

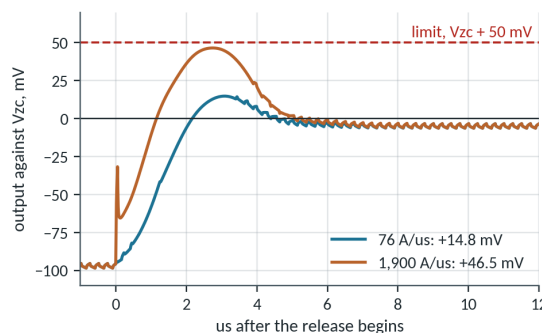
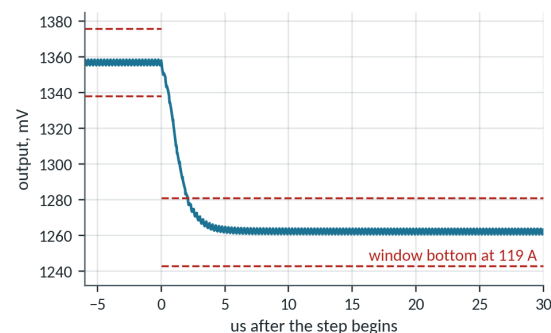


Key figures

Limits are Intel's for 04B or the engineer's two rules. Rows that do not name the TPS40091 were read with an ideal modulator.

Parameter	Conditions	Read in simulation	Limit
Output current	thermal design current 101 A: see the notes	0 to 119 A	Icc max 119 A
Load line	24 A to 119 A, as the guide reads it	0.9962 mOhm	1.00 mOhm
Output against the centre of the window	24 A; 119 A; nominal	-0.13 mV; +0.22 mV	within 19 mV
Ripple, peak	through 20 MHz, at the corners	1.75 mV	5 mV
Load step, margin over the bottom of the window	24 A to 119 A in 1.25 us; nominal; low corner	17.7 mV; 15.5 mV	stay above the bottom
Release overshoot against Vz _c	100 A to 5 A, three tolerances stacked; at 76, 200, 500 and 1,900 A/us	+14.8, +32.3, +40.4, +46.5 mV	+50 mV
Loop crossover	119 A; 24 A; sense path 3 % low, sense filter 10 % slow	192,650 Hz; 193,806 Hz	under 200,000 Hz
Phase margin	the same two decks	86.0 deg; 82.6 deg	
Change of VID, margin over the minimum line	1.6 V to 0.8375 V and back, at 5 A and 101 A	4.6 mV or more	stay above the line
Stage efficiency	24 A; 100 A; 119 A	93.5 %; 91.4 %; 90.5 %	
Board estimate of efficiency	119 A at 25 C; with the FETs at 125 C	86.8 %; about 85.1 %	
TPS40091: step, against the window bottom	typical part; reference 1 % low and no trim	+10.7 mV; -3.7 mV	stay above the bottom
TPS40091: release overshoot against Vz _c	typical part, same corner; 76 and 1,900 A/us	+23.8 mV; +47.8 mV	+50 mV
TPS40091: loop crossover, phase margin	typical part, 119 A	148,162 Hz, 75.2 deg	under 200,000 Hz
TPS40091: phase currents with sense offsets	3.5 mV either way; 119 A, even share 29.7 A	23.1 A and 36.4 A	

Two of the tests



Left: the guide's step, 24 A to 119 A in 1.25 us, with Intel's window before and after (spike242). Right: the release from 100 A to 5 A with three tolerances stacked, at the slowest and the fastest speed read (spike241, spike216). Both with the ideal modulator.

The power stage as simulated

High side CSD17578Q3A and low side CSD17573Q5B, one of each a phase, on the makers' models. Driver TPS28225 at 7.2 V with the part's timing. Inductor 150 nH, 0.29 mOhm (FP1007R3-R15-R). Sense resistor 0.5 mOhm, 1 %, through a gain of 20. Output capacitors 23 x CL32A107MQVNNNE, 68.0 uF a part at 1.3 V by the maker's model. With the ideal modulator the compensator has gain 2.5, a zero at 500 kHz and a pole at 1,040.88 kHz. Controller TPS40091, in the decks as a network built from its datasheet, with 8 kOhm and 1 nF on its amplifier.

Before a board

- One high-side FET a phase carries about 1.9 W at 101 A. Lay out two.
- The TPS40091 has no VID input, a 1 % reference and up to 3.5 mV of sense offset: a DAC outside it, a trim on each board, unequal phases.
- With the phases on one side of the socket, the capacitors stand at the socket.
- No deck carries protection, the input filter, the socket or the lands.